



MINING DEALER BEST PRACTICE

Improved Assembly Process of the 793D/F XL Front Brake

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

During the installation activity of the XL front brake assembly of the 793D or F after being rebuilt, it is common to find a misalignment between the splines of the discs and plates in relation to the brake assembly due to the lack of an appropriate tool for proper alignment.

Since this misalignment takes place during component PCR, it is necessary to fix it in the field upon the brake assembly removal, placing it horizontally for alignment between the parts before component installation in the suspension and front wheel set, ending with the bearings preload.

This problem significantly increases the time spent for the front brake assembly replacement process.

2.0 Best Practice Description

This best practice describes the development and use of a device/tool to mitigate the impact of two types of misalignments during the assembly process, as follows:

A. Disc spline misalignment:

Since Hub, PN 265-3189 is not always delivered from customer site to the CRC together with the brake assembly, it often leads to misalignment.



Figure 1 - Disc spline misalignment in relation to the 265-3189 hub during the component installation.

B. Decentralization of the plates and disks in relation to the hub:

As much as the hub is used, this will ensure that the splines are aligned during assembly (previous example), however the plates and disks can still be installed centered in relation to the brake assembly (housing).

The images below show a set of brake with the splines of the discs aligned but showing the decentralization in relation to the set (housing) which makes it impossible to install the Hub during the assembly of the component in the field.

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Figures 2 and 3 - Decentralization of the discs and plates in relation to the set.

Note: The idea contained in this best practice, written for 793F trucks can be replicated to all other truck models with XL brake arrangements, since you make the adjustments needed according to the specific hub used and dimensions of each one.

3.0 Implementation Steps

The customization of a Hub, PN 265-3189 to be used as a tool will make it possible to correct these two problems mentioned in the previous topic.

- Acquisition of a PN 265-3189 Hub:



Figures 4,5 and 6 - 265-3189 Hub drawings

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- Suitability Tool:

This tool can be adapted from the Hub, PN 265-3189 with the addition of spacers welded along the inner diameter of the part according to the specifications below:



Figures 7 and 8 - Spacers (four welded bosses)

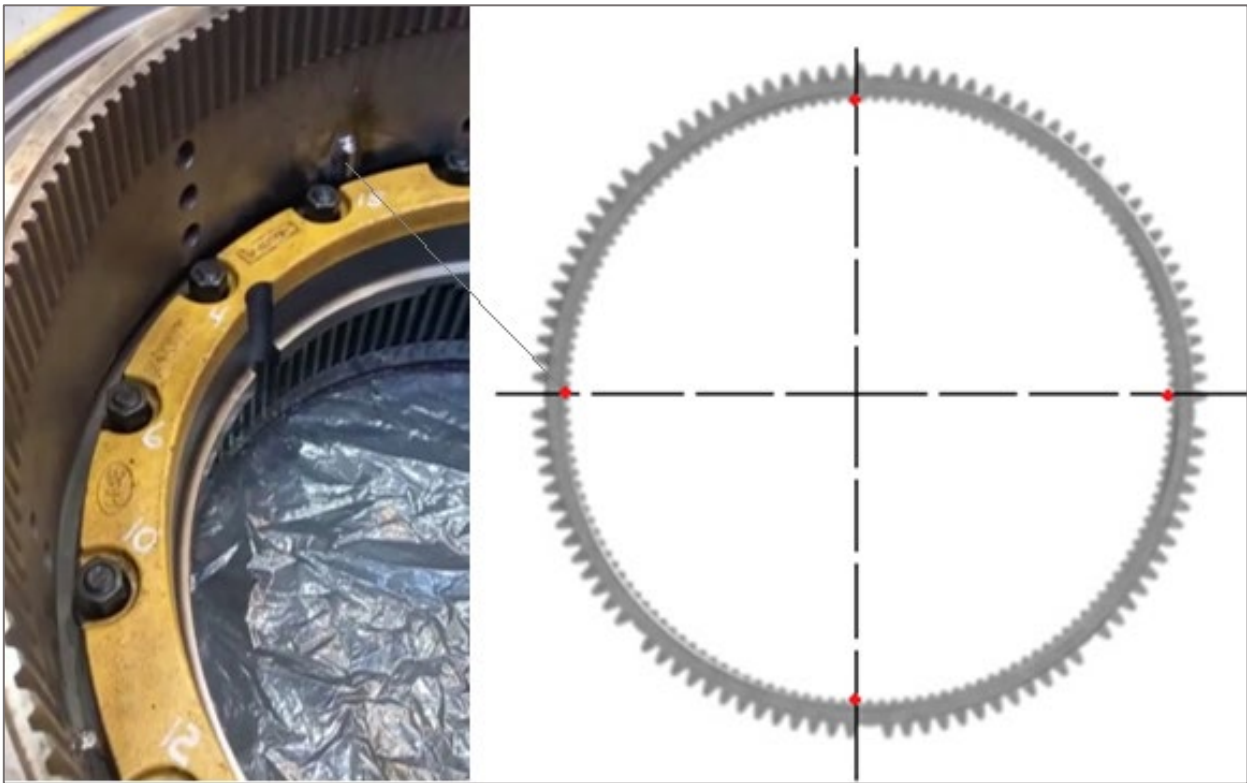


Figure 9 - 04 X 31,50mm of Steel bar with 11mm Ø (welding position).

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- Utilization procedure for the tooling:
 - 1 - Position the brake assembly in a horizontal position on a service bench;
 - 2 - Use an appropriate tool to release the parking brake in order to keep the plates and discs free to move;
 - 3 - Install an appropriate lifting device for the tooling (customized hub with welded spacers);
 - 4 - Position and slide the tooling between the splines of the brake discs;
 - 5 - Make sure the brake discs are correctly aligned and centered;
 - 6 - Apply the parking brake;
 - 7 - Remove the tooling from the brake assembly.

4.0 Benefits

- **Safety:** Eliminate the risk of injuries by avoiding rework in the field and preventing technician's exposure to the risks inherent to correlated activities.
- **Contamination control:** Once the component is received/ready for installation in the machine, there is no need for disassembling it thus eliminating exposure to external environmental contaminants.
- **Reliability of CRC service:** Use of the proposed tool for brake assembly alignment provides a fast, safe and quality installation; damage to the parts is avoided.
- **Elimination of rework during component installation:** This process ensures that the front brake assembly is installed on the machine in the standard time as it eliminates the possibility of field rework.

The chart below shows the installation time considering customer's standard time to carry out the R&I, that is, from the equipment stop in the maintenance box to the release of the machine for operation, including the tire repair and lubrication works.

The process without using this suitability tool (total time with rework +R&I) spends 10,5h (161%) more than optimized one (Optimized R&I).

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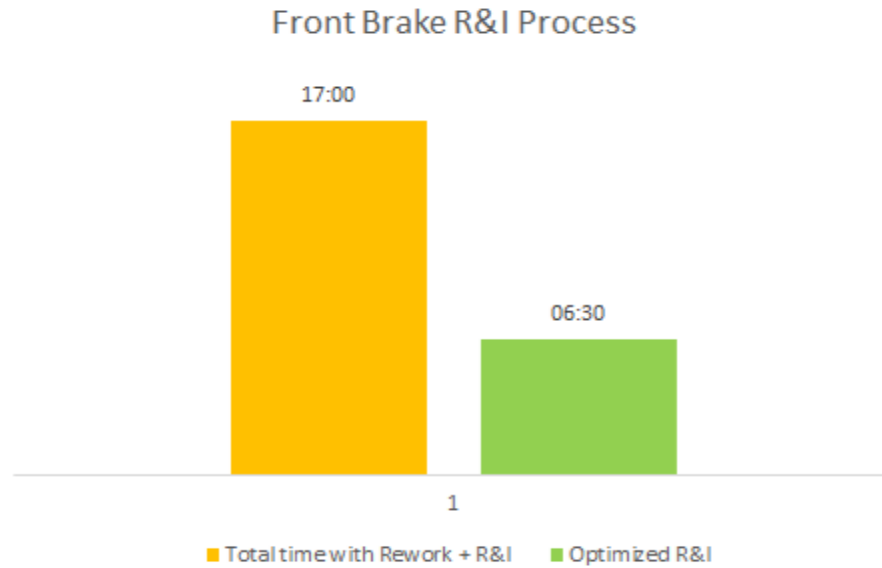


Chart 1 – Difference between the former process VS Optimized

5.0 Resources Required

- 265-3189 Hub;
- Welding process and supplies
- 04 X 31,50mm of Steel bar 11mm Ø.

6.0 Supporting Attachments / References

- Cleaning Front Wheel Groups Of 793, 795, And 797 Off-Highway Trucks {4002, 4006, 4008, 4200, 4205, 4208, 4215, 4234, 4251} (SEBF8490-03)
- Wheel, Brake And Suspension (Front) - Install - Extended Life (UENR2257-10)
- Service And Parking Brake (Front) - Remove And Install (UENR1142-04)
- Service And Parking Brake (Front) - Disassemble (UENR1142-04)
- Service And Parking Brake (Front) - Assemble (UENR1142-04)

7.0 Related Best Practices

This best practice may be used joint with the BP below:

- 1206-4.03-1277 - Disassemble / Assemble Tool for 793D/F Hub Brake

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